

# Curriculum Vitae (Jooyoung HAHN)

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## 1 Contact Information

Department of Software Engineering,

Faculty of Nuclear Sciences and Physical Engineering,

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## 2 Education

*Ph.D., Mathematical Sciences* ..... March 2002 - February 2008

- [Korea Advanced Institute of Science and Technology](#) (KAIST), Deajeon, South Korea
- Thesis title: **PDE-based image processing for segmentation and image restoration**
- Advisor: Professor [Chang-Ock Lee](#)
- Area of Study: Image processing based on nonlinear PDEs, variational numerical methods, nonlinear optimization

*M.S., Mathematics* ..... March 2000 - February 2002

- [KAIST](#), Deajeon, South Korea
- Advisor: Jin Hwan Lim
- Area of Study: Riemannian geometry

*B.S., Electronic Material & Devices Engineering* ..... March 1995 - February 2000

- Department name changed from Applied Physics
- [Inha University](#), Incheon, South Korea
- Second major: Mathematics

## 3 Professional Career

*Assistant Professor* ..... June 2025 - Present

- Research topics: Numerical PDEs on polyhedral meshes, Topological data analysis on biomedical and chemical data. Neural network approaches in PDEs

*SASPRO2 Fellow* ..... February 2022 - May 2025

- Funding: [SASPRO2](#) (Slovak Academic and Scientific Programme, Horizon 2020 Marie Skłodowska-Curie COFUND Action Programme)
- Hosting University: [Department of Mathematics and Descriptive Geometry, Faculty of Civil Engineering, Slovak University of Technology](#), Radlinskeho 11, 810 05 Bratislava, Slovakia
- Principal Investigator: PhD [Jooyoung Hahn](#)
- Science in Charge: Professor [Karol Mikula](#)
- Topic: Numerical methods for computational evolving manifolds

*Senior Development Engineer Software Fluid Dynamics* ..... May 2012 - December 2021

- Advanced Simulation Technologies, [AVL](#), Graz, Austria
- Computational fluid dynamics in advanced simulation technologies
- The main numerical algorithm developer in finite volume method and particle-based method on polyhedral meshes:
  - $G$ -equation (level set method) in combustion on polyhedral meshes.
  - Curve evolution for spark plug system (Particle movement controls in a complicated geometry).

- Efficient surface-to-surface radiation.
- Level set method with adaptive mesh refinement for multiphase flow.
- Accurate gradient computations

*Rapid Innovation Team, Advanced Simulation Technologies, AVL* ..... Mar 2019 - June 2019

- Advanced Simulation Technologies, [AVL](#), Graz, Austria
- Complete Vehicle Energy Management Control Optimization (Level-Set Dynamic Programming)

*K-TAG Europe* ..... July 2016 - Present

- About Korea-Technology Advisory Group (K-TAG) Europe : K-TAG Europe launched by [KIAT](#) and MOTIE in July 2014, consists of Korean engineering experts in Europe. Main activities of K-TAG are (1) to assist Korean Small and Medium-sized Enterprises (SMEs) in finding European Innovative partners (2) to provide advice as well as information related to Korea-EU R&D cooperation (3) to develop and participate in Korea-EU joint R&D projects.

*EUREKA/Eurostars: Technical expert* ..... July 2019 - Present

*Senior Postdoc.* ..... November 2010 - April 2012

- [Institute for mathematics and scientific computing](#), University of Graz, Graz, Austria
- Advisor: Professor [Michael Hintermüller](#)
- A senior PostDoc in [START Project: Interfaces and Free Boundaries](#)

*Research staff* ..... November 2008 - October 2010

- [Division of Mathematical Sciences](#), Nanyang Technological University, Singapore
- Advisor: Professor [Xue-Cheng Tai](#)
- A research staff in [Mathematical Imaging and Vision Group](#)

*Postdoc.* ..... March 2008 - October 2008

- [Department of Mathematical Sciences KAIST](#), Deajeon, South Korea
- Advisor: Professor [Chang-Ock Lee](#)
- A PostDoc in [Computational Mathematics and Imaging Lab.](#)

*Visiting researcher* ..... August 2005 - July 2006

- [Institute of mathematics and its applications](#) (IMA), Minneapolis, Minnesota, USA
- Solved a challenge problem posted in [IMA Impacts: An eye for aphids](#)
- Funding: Korean Research Foundation, KRF-2006-311-C00015.

*Research assistant* ..... January 2005 - March 2005

- Nourishment of KAIST brand supported by the small and medium business administration & KAIST
- Cooperated with [Interactive technology for the value improvement](#) (INTVIM)
- Result: Development of 3D virtual reality engine based on real photo

## 4 Academic Experience

### *Teaching*

- CTU in Prague ..... September 2025 - Present
  1. Basics of Programming, September 2026 - December 2026
  2. Basics of Algorithmization, February 2026 - May 2026
- STUBA ..... February 2022 - December 2024
  1. Coding project in data analysis: Can we see wine in 2D?, September 2024 - December 2024
  2. English: Professional presentation in applied mathematics, September 2024 - December 2024
  3. English: Level set method in industrial problems, February 2024 - May 2024
  4. English: Professional presentation in applied mathematics, September 2023 - December 2023
  5. English: Seminal papers in image processing before AI era, February 2023 - May 2023
  6. Finite volume method: Coding in [AVL FIRE™](#) for the evolution of surfaces on polyhedral meshes, September 2022 - December 2022

- 7. English: Professional presentation in applied mathematics, September 2022 - December 2022
- 8. Numerical analysis of differentiation equations: Level Set Methods in Evolving Manifold, February 2022 - May 2022
- KAIST ..... March 2008 - May 2008
- 1. Numerical Analysis I

*Computer system manager* ..... March 2002 - August 2003

- [Computational mathematics laboratory](#), KAIST
- Regular system management of Linux and Window XP
- Establishment of integrated Window XP system on Linux using Samba primary domain controller

*Teaching assistant*

- Calculus, Linear algebra, Set theory, Ordinary differential equation, Differential geometry, KAIST, South Korea, March 2000 - February 2004
- Calculus, NTU, Singapore, November 2008 - October 2010

*Undergraduate researcher* ..... March 1997 - February 1998

- [Information display laboratory](#), Electronic engineering, Inha university
- Study of tensor fields for understanding physical properties of liquid crystal display

*Undergraduate researcher* ..... March 1996 - February 1997

- [Photonic integrated circuit & quantum device laboratory](#), Information & communication engineering, Inha university
- Construction of small model of HESS (Hoop energy storage system)

## 5 Invited Talks

- Humanoid Olfactory Display International Research Conference, Busan, South Korea, June 10-13, 2026, Title: *From High-Dimensional Chaos to Shape: Topological Data Analysis for Untargeted Volatile Profiling and Tumor Microenvironment*
- Data Science Convergence, Busan, South Korea, December 17-19, 2025, Title: *A Topological Data Analysis Framework for Malignant-Immune Spatial Analysis*
- AI-CFD International Mini Symposium Scientific Machine Learning and AI for CFE, POSCO Center, Seoul, South Korea, August 21, 2025, Title: *Implicit Neural Representations for Partial Differential Equations.*
- (Seminar) IBS Biomedical Mathematics Group (BIMAG), Daejeon, South Korea, August 6, 2025, Title: *Topological Data Analysis with two applications: Tumor Microenvironment and 2D Chromatography with High-Resolution Mass Spectrometry*  
[Recording: YouTube](#)
- Emerging Trends in Computational Fluid Dynamics: Towards Industrial Applications, Stanford University Redwood City, CA 94063, United States, November 5-7, 2024, *Numerical methods for computational evolving manifolds in industrial applications.*  
[Recording: CASSYNI](#)
- Advances in High-order Methods - Fluid Dynamics, Biomedical Science, and Exascale Computing, PostTech, Pohang, South Korea, June 10-13, 2024, Title: *Level set methods on polyhedral meshes.*
- (Seminar) A series of seminars, KAIST, Daejeon, South Korea, July 19-21, 2023, Title: *Image Processing before the Era of Neural Network.*
- Discrete Duality Finite Volume Method and Applications, Jean-Morlet Chair - Workshop, CIRM, Marseille, France, October 17-21, 2022, *Finite Volume Method for Level Set Equations on Polyhedral Meshes.*
- Workshop on Scientific Computing 2022, Dčín, Czech Republic, May 26-29, 2022, Title: *What is a proper boundary condition to solve eikonal equation on a non-convex domain?*
- ALGORITHM20, Conference on Scientific Computing, Vysoke Tatry, Podbanske, September 10-15, 2020, Title: *Finite volume level set methods in combustion engines*
- Summer School for modeling in medical mathematics, July 2-4, 2019, Title: *Modeling for Medical Mathematics: Evolution of Manifolds, Finite Volume Method, and Deep Learning*

## 6 Minisymposium Organizer

- WCCM/ECCOMAS, July 19-24, 2026, Munich, Germany, *Title: Neural Network Solvers for PDEs: Bridging Theory and Practice in Scientific Computing - Parts I-V*, with Alexander Heinlein (TU Delft)
- ACOMEN2025, September 15-19, 2025, Ghent, Belgium, *Title: Advancements in Neural Network Solvers for PDEs: Applications and Challenges*, with Minseok Choi (POSTECH)
- SIAM/CAIMS Annual Meetings, July 28 - Aug 1, 2025, Montréal, Canada, *Title: Advances in Neural Network Methods for PDEs and Optimization - Parts I, II, and III*
- SIAM Conference on Imaging Science, May 28-31, 2024, Atlanta, USA, *Title: Surface reconstruction: PDEs, Variational, and Deep Learning Methods - Part I and II*
- ALGORITHMY 2024, March 15-20, 2024, High Tatras Mountains, Slovakia, *Title: Numerical methods for level-set and eikonal equations - theory and applications*
- International Council on Industrial and Applied Mathematics, August 20-25, 2023, Tokyo, Japan, *Title: Numerical Algorithms for the Eikonal Equation and Its Applications*
- International Council on Industrial and Applied Mathematics, July 15-19, 2019, Valencia, Spain, *Title: Advanced numerical methods for evolving manifolds*
- Europe-Korea Conference on Science and Technology, August 20-24, 2018, Glasgow, UK. *Title: Information technology and machine learning*
- 6<sup>th</sup> European Conference on Computational Mechanics, 7<sup>th</sup> European Conference on Computational Fluid Dynamics, June 11-15, 2018, Glasgow, UK. *Title: Moving interface problems in computational fluid dynamics*
- Europe-Korea Conference on Science and Technology, July 26-29, 2017, Stockholm, Sweden. *Title: Industrial mathematics for scientific computing: Imaging and machine learning and computational fluid dynamics*
- Europe-Korea Conference on Science and Technology, July 27-30, 2016, Berlin, Germany. *Title: Contemporary topics in image processing and computer vision*
- SIAM Imaging Science, May 23-26, 2016, New Mexico, USA. *Title: PDE-based image processing: Reconstruction, filtering, segmentation, compression, and inpainting*
- International Conference on Numerical Combustion, April 19-22, 2015, Avignon, France. *Title: Surface evolution methods in gasoline direct injection combustion engines*
- SIAM Imaging Science, May 20-22, 2012, Philadelphia, USA. *Title: Functional analysis and accurate numerical methods in image processing*
- International Council on Industrial and Applied Mathematics, July 18-22, 2011, Vancouver, Canada. *Title: Fast optimization algorithms in image processing and its applications*
- SIAM Imaging Science, April 12-14, 2010, Chicago, USA. *Title: Surface reconstruction from sparse gradient fields, unorganized point clouds, and photometric stereo*

## 7 Patent

- Real-time fine dust monitoring system and its methods, KR20210050831A, 2021, Seungjun Jeon, Jun Beum Kim, Jooyoung Hahn, and Seong Weon Jeong.
- A method for reconstructing a 3d surface from a 2d sketch, CN103460252B, 2017, Jie Qiu, Jooyoung Hahn, Eiji Sugisaki, Lei Jia, Xue-Cheng Tai, and Hock Soon Seah.
- Methods and systems for generating enhanced images using Euler's Elastica model, US8447135B2, 2013, Xue-Cheng Tai, Jooyoung Hahn, Jason Ginmo Chung.
- A method for reconstructing a 3d surface from a 2d sketch, WO2012118439A1, 2012, Jie Qiu, Jooyoung Hahn, Eiji Sugisaki, Lei Jia, Xue-Cheng Tai, and Hock Soon Seah.
- Method for image segmentation using statistically reinstating force and multi-resolution, KR100925180B1, 2009, Chang-Ock Lee, Sung Ha Park, and Jooyoung Hahn.

## 8 Funding

AVL AST University Partnership Program (principal investigator)

- In-kind sponsorship in the form of software licenses.
- AVL FIRE™ and PREONLAB, May 2022 - Apr 2026. Accepted.

Global Visiting Fellowship at Seoul National University, South Korea

- Topic: PDE-Based Classification for High-Dimensional, Sparsely Annotated Data. July 2026 - September 2026. Accepted.
- Topic: Operator Learning of Grad-Shafranov Equation. August 2025 - September 2025. Accepted.
- Topic: Application of the eikonal equation and Surface-to-surface radiation. June 2024 - August 2024. Accepted.
- Topic: Neural network solver for PDEs. June 2023 - August 2023. Accepted.

*SASPRO2/Horizon 2020 Marie Skłodowska-Curie COFUND Action Programme (principal investigator)*

- Title: Numerical methods for computational evolving manifolds, Feb 2022 - Jan 2025. Accepted.

*K-TAG: Technical Consulting for EUREKA/EUROSTAR*

- Title: Analysis and prediction of evolving particulated matter in the air, 2018, [Metariver Technology Co., Ltd.](#). Accepted.
- Title: Health care system for emotion and stress measurement based on image processing, 2019, [Inforshare](#). Accepted.
- Title: Exploration of Overseas Partners and International Collaborative R&D Project Planning for Real-time Simulation Cyber-Physical Systems (CPS) System Development, 2020, [UVC Co.,Ltd.](#). Accepted.

*EIC Pathfinder (co-investigator)*

- Title: Unified Diagnosis Platform by Nano-Metric Molecular Sensing, Submitted in 2022 and 2023. Not accepted.

*Industrial Project (project leader in AVL)*

- Title: Finite volume level-set methods for  $G$ -equation in AVL FIRE™, Feb 2017 - Jul 2017. Accepted.

*FETOPEN (co-investigator)*

- Title: Computational Evolving Manifolds for Future Emerging Technology, Submitted in 2015 and 2016. Not accepted.

*Humbolt Research Fellowship in 2012*

- Title: Higher-order Regularity in PDE-based Image Analysis and Computer Vision, Host: Prof. Daniel Cremers in Department of Computer Science, Technical University of Munich, Germany. Not accepted.

*Korean Research Foundation, KRF-2006-311-C00015*

- Visiting researcher at [Institute of mathematics and its applications](#) (IMA), Minneapolis, Minnesota, USA, Aug 2005 - Jul 2006. Accepted

## 9 Awards

*K-TAG awards 2019 for consulting business*

- Analysis and prediction of evolving particulated matter in the air
- Health care system for emotion and stress measurement based on image processing.
- News: [click the link](#)

*English Teaching Award: Numerical Analysis I, KAIST, Spring semester in 2008*

*KSIAM Young researcher award*

- [Korean society for industrial and applied mathematics](#) (KSIAM) 2007 spring conference, May 25-26, 2007, KAIST, South Korea
- Title: Fine segmentation using geometric attraction-driven flow and edge-regions

*University students contest of mathematics held by the Korean Mathematical Society*

- Honorable mention prize in November 1998
- Bronze prize in November 1997

*Inha University*

- ***Summa cum Laude***, with honors in engineering
- University scholarship, 1995-1999

## 10 Languages

*Communication*

- English (Fluent)
- Korean (Native)
- German (Basic)

*Computation*

- Python (2022 - present): Development of data analysis in 2D chromatography, tumor microenvironments, and high performance computing in TDA.
- FORTRAN (2012 - present): Development of commercial CFD software and computational geometry algorithms.
- MPI and parallel computing (2012 - present): Development of commercial CFD software and computational geometry algorithms.
- Linux (2002 - present), System manager of Linux (2002 - 2003): Integration of linux and window systems, security, web server, etc.
- C, C++ (1995 - 2014), Visual Studio, MFC, API (2002 - 2008): Development of in-house codes for computer vision and image processing. (three patents)
- GPU (2008 - 2010): Development of real-time 3D reconstruction from hand drawing sketch. (a patent)
- Others: Tools for TDA (Gudhi and Ball Mapper), Latex, Matlab, HTML

## 11 Soft Skills

*Singing in a choir: 2<sup>nd</sup> Bass*

- [UniChor in Graz](#) by the conductor Matthias Unterkofler
- [Cappella Nova Graz](#) by the conductor Otto Kargl

*Cooking*

- [Home Cooking \(Instagram\)](#)